

Lab 4
Library

TA: Yu-Wen Huang

Prof.: Hsung-Pin Chang

NCHU System & Network Lab

$ Outline

- Introduction
- Static Library
- Dynamic Library

$ Introduction (1/3)

- **Library**
  - 將一些行為 **(behavior)** 實作而成的集合 **(collection)**
  - 定義了良好的介面 **(interface)**，按照指示呼叫即可
- 常見的例子
  - 不用定義就可以呼叫 **printf()**
  - 用 **high-level** 的程式語言實作 **system call**

$ Introduction (2/3)

- Library 主要分成兩大類
  - **Static Library** (aka archives)
  - **Dynamic Library** (aka shared library)

特性	Static Library (.a, .lib)	Dynamic/Shared Library (.so, .dll, .dylib)
連結方式	編譯時直接複製進程式	執行時載入
檔案大小	大（程式包含所有 lib code)	小（只含符號表）
更新方式	要重新編譯程式	替換 shared lib 即可
執行效率	較快（少一層間接呼叫）	稍慢（需動態載入）
共用性	無	可被多程式共用

$ Introduction (3/3)

- Example

```
// main.c
#include<stdio.h>
#include "lib.h"

int main() {
    int n1 = add(5);
    int n2 = sub(5);
    printf("After add, n1 = %d\n",
n1);
    printf("After sub, n2 = %d\n",
n2);

    return 0;
}
```

```
// lib.h
int add(int
arg);
int sub(int
arg);
```

```
// add.c
#include "lib.h"
int add(int arg) {
    arg++;
    return arg;
}
```

```
// sub.c
#include "lib.h"
int sub(int arg) {
    arg--;
    return arg;
}
```

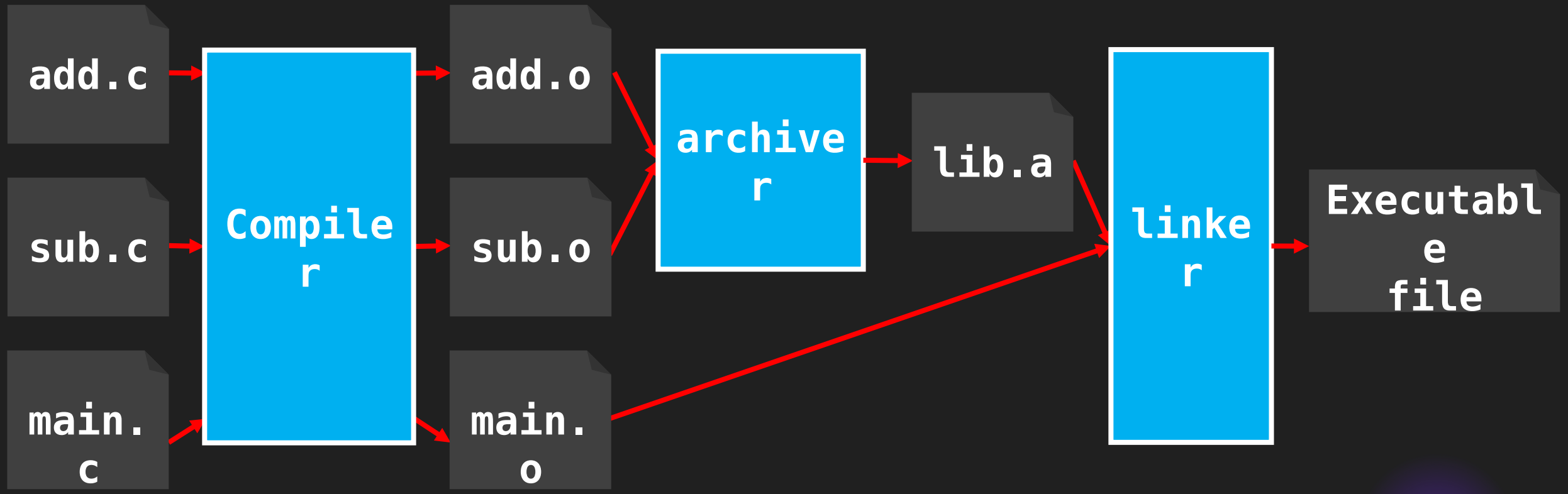
$ Outline

- Introduction
- Static Library
  - Definition
  - Create a Static Library
  - Exercise 1
- Dynamic Library

$ Definition (1/2)

- 在編譯時期就執行 **link** 的動作
- **binary program** 直接包含了 **object files** 和 **libraries**
  link
- 優點：不用考慮相依性問題
- 缺點
  - 若許多程式使用同一個 **library**，將會浪費硬碟和記憶體空間
  - 如果要更新 **library**，需要重新編譯整支程式

$ Definition (2/2)



$ Outline

- Introduction
- Static Library
  - Definition
  - Create a Static Library
  - Exercise 1
- Dynamic Library

$ Create a Static Library (1/3)

- Command "ar"

```
oslab-VirtualBox:~/oslab/lab4$ ar
Usage: ar [emulation options] [-]{dmpqrstx}[abcDfilMNoOPsSTuvV] [--plugin <name>] [member-name] [count] archive-file file...
       ar -M [<mri-script]
 commands:
  d            - delete file(s) from the archive
  m[ab]        - move file(s) in the archive
  p            - print file(s) found in the archive
  q[f]         - quick append file(s) to the archive
  r[ab][f][u]  - replace existing or insert new file(s) into the archive
  s            - act as ranlib
  t[O][v]      - display contents of the archive
  x[o]         - extract file(s) from the archive
 command specific modifiers:
  [a]          - put file(s) after [member-name]
  [b]          - put file(s) before [member-name] (same as [i])
  [D]          - use zero for timestamps and uids/gids (default)
  [U]          - use actual timestamps and uids/gids
  [N]          - use instance [count] of name
  [f]          - truncate inserted file names
  [P]          - use full path names when matching
  [o]          - preserve original dates
  [O]          - display offsets of files in the archive
  [u]          - only replace files that are newer than current archive contents
 generic modifiers:
  [c]          - do not warn if the library had to be created
  [s]          - create an archive index (cf. ranlib)
  [l <text> ]  - specify the dependencies of this library
  [S]          - do not build a symbol table
  [T]          - deprecated, use --thin instead
  [v]          - be verbose
  [V]          - display the version number
  @<file>      - read options from <file>
  --target=BFDNAME - specify the target object format as BFDNAME
  --output=DIRNAME - specify the output directory for extraction operations
  --record-libdeps=<text> - specify the dependencies of this library
  --thin       - make a thin archive
```

$ Create a Static Library (2/3)

- 我們使用前面 example 的程式碼，輸入以下指令
  - **gcc –c add.c sub.c**
  - **ar rcs lib.a add.o sub.o**
    - **r = replace**
    - **c = create**
    - **s = symbol index**
  - **gcc main.c –L. lib.a –o example_static**
    - **-L. 告訴 compiler 去當前目錄找 library**

$ Create a Static Library (3/3)

```
oslab-VirtualBox:~/oslab/lab4/example$ ls
add.c  lib.h  main.c  sub.c
oslab-VirtualBox:~/oslab/lab4/example$ gcc -c add.c sub.c
oslab-VirtualBox:~/oslab/lab4/example$ ls
add.c  add.o  lib.h  main.c  sub.c  sub.o
oslab-VirtualBox:~/oslab/lab4/example$ ar rcs lib.a add.o sub.o
oslab-VirtualBox:~/oslab/lab4/example$ ls
add.c  add.o  lib.a  lib.h  main.c  sub.c  sub.o
oslab-VirtualBox:~/oslab/lab4/example$ gcc main.c -L. lib.a -o example_static
oslab-VirtualBox:~/oslab/lab4/example$ ls
add.c  add.o  example_static  lib.a  lib.h  main.c  sub.c  sub.o
oslab-VirtualBox:~/oslab/lab4/example$ ./example_static
After add, n1 = 6
After sub, n2 = 4
oslab-VirtualBox:~/oslab/lab4/example$
```

$ Outline

- Introduction
- Static Library
  - Definition
  - Create a Static Library
  - Exercise 1
- Dynamic Library

$ Exercise 1 (1/2)

- 請撰寫兩支程式
  - **sort.c**
    - 建立 **sort** 函式，參數可以接受陣列的位址
    - 將陣列內的整數從小排到大
  - **exercise.c**
    - 建立一個長度為 **5** 的陣列，放入 **5** 個不重複隨機整數進去
    - 先印出 **sort** 之前的 **5** 個整數
    - 呼叫 **sort** 函式，傳入該陣列的位址，並印出 **sort** 完的結果

$ Exercise 1 (2/2)

- 將 **sort.o** 加入至 **exlib.a**
- 最後使用 **gcc** 將 **exercise.c** 與 **exlib.a** 輸出成 **exercise1**
- 現場 Demo 如下執行一次即可，做不完也可以回家截圖並上傳至 **iLearning**

```
oslab-VirtualBox:~/oslab/lab4/exercise$ gcc -c sort.c
oslab-VirtualBox:~/oslab/lab4/exercise$ ar rcs exlib.a sort.o
oslab-VirtualBox:~/oslab/lab4/exercise$ gcc exercise.c -L. exlib.a -o exercise1
oslab-VirtualBox:~/oslab/lab4/exercise$ ./exercise1
Before sort: 65 33 37 5 101
After sort: 5 33 37 65 101
oslab-VirtualBox:~/oslab/lab4/exercise$ ./exercise1
Before sort: 101 69 37 97 1
After sort: 1 37 69 97 101
oslab-VirtualBox:~/oslab/lab4/exercise$
```

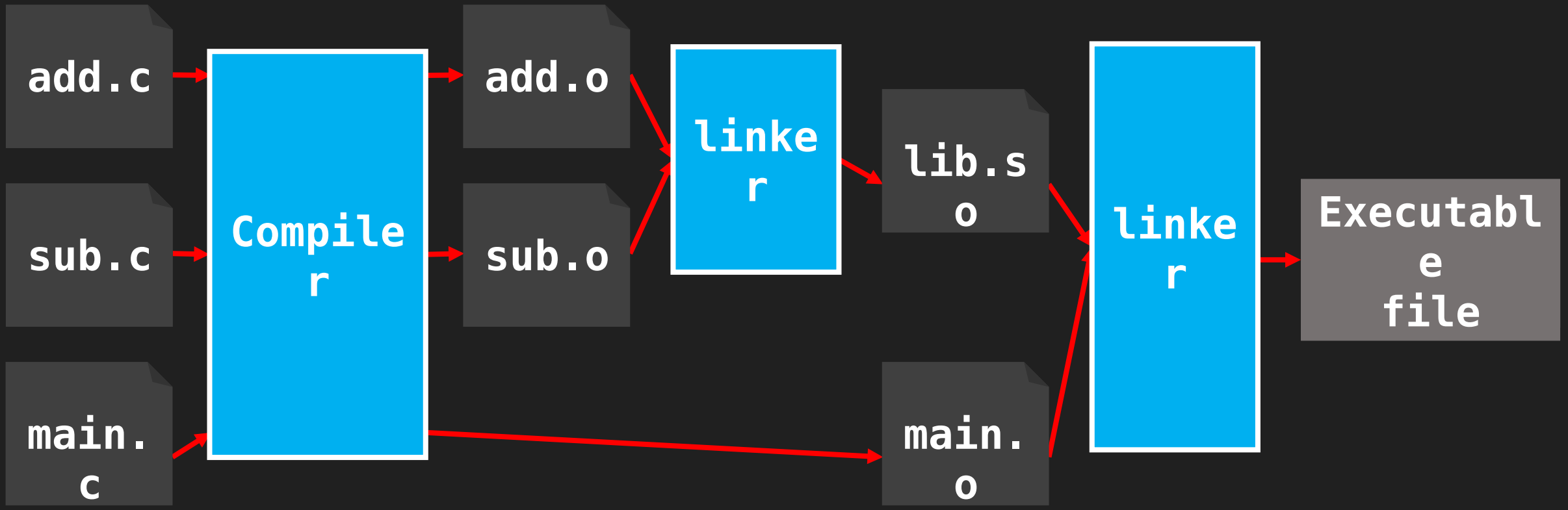
$ Outline

- Introduction
- Static Library
- Dynamic Library
  - Definition
  - Create a Dynamic Library
  - Exercise 1

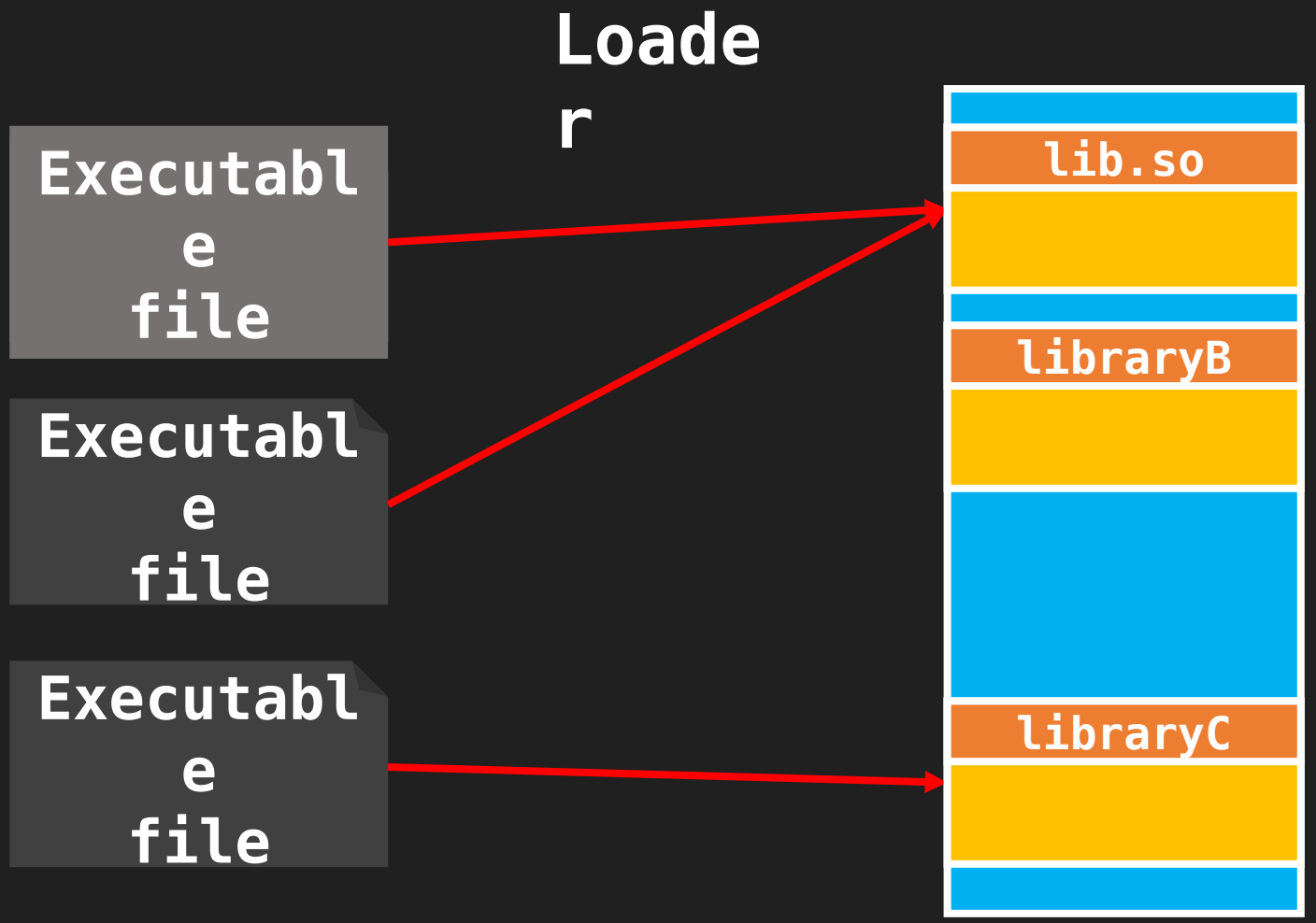
$ Definition (1/3)

- 將 **library** 載入到記憶體中
- 需要該 **library** 的程式會等到 **execution time** 再連結
- 優點：節省較多硬碟和記憶體空間
- 缺點：若不同程式所需的版本不一，可能會有相依性問題

$ Definition (2/3)



$ Definition (3/3)



$ Outline

- Introduction
- Static Library
- Dynamic Library
  - Definition
  - Create a Dynamic Library
  - Exercise 2

$ Create a Dynamic Library (1/4)

- 我們使用前面 example 的程式碼，輸入以下指令
  - **gcc –c –fPIC add.c sub.c**
    - **-fPIC: generate position-independent code**
  - **gcc add.o sub.o -shared –o lib.so**
  - **gcc main.c –L. lib.so –o example_dynamic**

$ Create a Dynamic Library (2/4)

- 但是執行的時候，卻回報說找不到 **library**
- 原因是我們的 **library** 沒有在 **dynamic linker** 清單中
- 而絕大部分的 **shared library** 都放在 **/lib** 或 **/usr/lib** 中

```
oslab-VirtualBox:~/oslab/lab4/example$ gcc -c -fPIC add.c sub.c
oslab-VirtualBox:~/oslab/lab4/example$ gcc add.o sub.o -shared -o lib.so
oslab-VirtualBox:~/oslab/lab4/example$ gcc main.c -L. lib.so -o example_dynamic
oslab-VirtualBox:~/oslab/lab4/example$ ./example_dynamic
./example_dynamic: error while loading shared libraries: lib.so: cannot open shared object file: No such file or directory
oslab-VirtualBox:~/oslab/lab4/example$
```

$ Create a Dynamic Library (3/4)

- 因此只要複製 **lib.so** 到 **/lib** 即可

```
oslab-VirtualBox:~/oslab/lab4/example$ sudo cp lib.so /lib/lib.so
oslab-VirtualBox:~/oslab/lab4/example$ ./example_dynamic
After add, n1 = 6
After sub, n2 = 4
oslab-VirtualBox:~/oslab/lab4/example$
```

$ Create a Dynamic Library (4/4)

- **Command "ldd": 列出該程式使用到的 share library**

```
oslab-VirtualBox:~/oslab/lab4/example$ ldd example_dynamic
        linux-vdso.so.1 (0x000070ac59624000)
        lib.so => /lib/lib.so (0x000070ac59605000)
        libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x000070ac59200000)
        /lib64/ld-linux-x86-64.so.2 (0x000070ac59626000)
```

$ Outline

- Introduction
- Static Library
- Dynamic Library
  - Definition
  - Create a Dynamic Library
  - Exercise 2

$ Exercise 2 (1/2)

- 請使用 **Exercise 1** 的兩支程式，並使用 **shared library** 的方式編譯
- 成功後，請使用 **ls –l** 指令比較檔案的大小
- 現場 Demo 如下頁執行一次即可，做不完也可以回家截圖並上傳至 **iLearning**

$ Exercise 2 (2/2)

```
oslab-VirtualBox:~/oslab/lab4/exercise$ gcc -c -fPIC sort.c
oslab-VirtualBox:~/oslab/lab4/exercise$ gcc sort.o -shared -o exlib.so
oslab-VirtualBox:~/oslab/lab4/exercise$ gcc exercise.c -L.. exlib.so -o exercise2
oslab-VirtualBox:~/oslab/lab4/exercise$ sudo cp exlib.so /lib/exlib.so
oslab-VirtualBox:~/oslab/lab4/exercise$ ./exercise2
Before sort: 5 33 97 65 101
After sort: 5 33 65 97 101
oslab-VirtualBox:~/oslab/lab4/exercise$ ./exercise2
Before sort: 33 1 5 37 69
After sort: 1 5 33 37 69
oslab-VirtualBox:~/oslab/lab4/exercise$ ls -l
total 68
-rwxrwxr-x 1 brian20020109 brian20020109 16224 Aug 25 17:02 exercise1
-rwxrwxr-x 1 brian20020109 brian20020109 16200 Aug 25 17:05 exercise2
-rw-rw-r-- 1 brian20020109 brian20020109   580 Aug 25 17:02 exercise.c
-rw-rw-r-- 1 brian20020109 brian20020109  1614 Aug 25 17:02 exlib.a
-rw-rw-r-- 1 brian20020109 brian20020109    33 Aug 25 16:32 exlib.h
-rwxrwxr-x 1 brian20020109 brian20020109 15104 Aug 25 17:05 exlib.so
-rw-rw-r-- 1 brian20020109 brian20020109   348 Aug 25 16:32 sort.c
-rw-rw-r-- 1 brian20020109 brian20020109  1472 Aug 25 17:05 sort.o
oslab-VirtualBox:~/oslab/lab4/exercise$
```



Thanks for your listening